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PALEOMAGNETIC RESULTS FROM THE WAIPAPA TERRANE, NORTHLAND PENINSULA, NORTH ISLAND, NEW ZEALAND: TECTONIC IMPLICATIONS FROM WIDESPREAD REMAGNETIZATIONS

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Abstract. Paleomagnetic samples were collected from a wide variety of rock types of the Waipapa terrane, which is exposed along the eastern coast of the Northland peninsula. Twelve sites yield stable magnetic directions, which fail both a fold and regional attitude test at the 99% confidence level, and indicate that widespread remagnetization has occurred over a large (<200 km) area. Rock magnetic experiments suggest that the mechanism for remagnetization was through the formation of fine-grained magnetite and hematite and was probably controlled by local variation in fluid migration. The mean in situ direction for the 12 sites is I = -70.2°, D=14.4° (α95=7.1°, kappa=38.6), which is significantly different from the predicted axial dipole field directions at the 95% confidence level. The age of the remagnetization is unclear. In an effort to date the remagnetization, a synthetic apparent polar wander (APW) path was constructed for New Zealand using Australian data, and the apparent south pole from the remagnetized Waipapa sites (latitude=69.7°S, longitude=150.7°E, α95=11.2°) was compared to this APW path. Using this method, the age of the remagnetization is constrained to be between ≈12 and 58 Ma. A major change in the tectonic regime occurred at ≈25 Ma, which is evidenced by obduction of the Northland allochthon and initiation of arc volcanism and subduction. Fluid migration and/or elevated temperatures driven by the initiation of subduction and arc volcanism, along with the obduction of the Northland allochthon seem likely mechanisms for the remagnetization.

INTRODUCTION

The pre-Tertiary rocks of New Zealand form a broad belt which is thought to record part of the pre-Cenozoic history of the Pacific continental margin of Gondwana (Figure 1). These rocks have been divided into many distinct tectonostratigraphic terranes [e.g. Bishop et al., 1985]. Terranes of the Western Province are mostly lower to middle Paleozoic crystalline and sedimentary rocks together with minor Mesozoic volcanic rocks and voluminous Cretaceous arc-related intrusions [Tulloch, 1983; Mattinson et al., 1986]. These are thought to represent part of the late Paleozoic and Mesozoic margins of Gondwana [Howell, 1980; MacKinnon, 1983]. Terranes east of the Mesozoic suture referred to as the Median Tectonic Line (MTL)[Landis and Coombs, 1967] consist of arc, forearc, and accretionary complex related rocks. The relationship between all of these terranes remains unclear, although some evidence suggests that some of these terranes may be allochthonous to the New Zealand margin of Gondwana [Howell, 1980; Sporli and Gregory, 1981; Haston et al., 1989]. All of these terranes were assembled by the Early Cretaceous [Bishop et al., 1985].

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Seafloor spreading rifted the New Zealand microcontinent away from Australia and Antarctica during the Late Cretaceous and Paleocene [Molnar et al., 1975; Weissel et al., 1977; Weissel and Hayes, 1977]. During this time, and up until the latest Oligocene, thick sequences of passive margin sediments were deposited throughout New Zealand. In the latest Oligocene/earliest Miocene, the relative motion between the Pacific and Indian plates changed, causing the passive margin setting to be replaced by a compressional tectonic setting. A profound latest Oligocene/earliest Miocene unconformity is present throughout the North Island; this marks the beginning of the still ongoing Kaikoura orogeny. During this time, subduction initiated on the eastern coast of the North Island, the Northland allochthon was accreted, and the beginning of the present tectonic regime was established.

The Waipapa terrane is exposed exclusively on the North Island (Figure 2). The Caples-Pelorus terrane of the South Island is thought to be an equivalent or analog of the Waipapa terrane [Sporli, 1987]. The Waipapa terrane consists of mainly volcanic-derived terrigenous clastic rocks which are thought to have been deposited in a forearc setting. The terrane has been further subdivided into two facies [Kear, 1971]: the more proximal Morrinsville facies and the distal Hunua facies which contains disrupted layers of chert, spilite, limestones, and argillites.

The spilite-chert ± limestone associations of the Hunua facies are commonly found within zones of broken formation and melange. Frequently, the terrigenous sediments overlie the spilite-chert ± limestone assemblages in depositional contact. However, in all cases where the spilite-chert ± limestone assemblages are on top of the terrigenous sediments, the contact is tectonic [Sporli and Gregory, 1986; Sporli et al., 1989]. Further, age reversals have been found in sections separated by melange zones [e.g., Sporli et al., 1989]. These observations suggest that the spilite-chert ± limestone associations were the basement upon which the terrigenous sediments were deposited. These basement rocks were subsequently accreted into imbricated sheets. These basement rocks have been interpreted to represent old oceanic crust or seamounts [Sporli and Bell, 1976; Sporli, 1978].

Sparse fossils found within the terrigenous sedimentary rocks are mostly Jurassic in age [Sporli and Grant-Mackie, 1976; Sporli et al., 1989], although some Triassic fossils have recently been identified [Sporli et al., 1989]. These fossils are in contrast to Permian fusulinid limestones with Tethyan affinities found in the basement rocks of the Waipapa terrane [Sporli and Gregory, 1981]. Sporli and Gregory [1981] suggested that these fusulinid limestones may have originated at low latitudes and were later accreted to the New Zealand margin.

The original goal of our study was to use paleomagnetic data to determine paleolatitude for the pieces of oceanic crust in the Waipapa terrane, in an effort to understand the nature of the Paleozoic and Mesozoic accretionary history for New Zealand, and to test whether they were allochthonous to the margin of Gondwana as suggested by the fusulinid limestones [Sporli and Gregory, 1981]. The presence of widespread remagnetizations has precluded this original goal. However, understanding the timing and mechanism of the remagnetization may provide important clues into the more recent history for Northland, New Zealand.

PALEOMAGNETIC RESULTS

A wide variety of rocks types was collected along the eastern coast of the Northland peninsula where the Waipapa terrane is well exposed. Table 1 lists a description of each of these sites; their locations are shown in Figure 2. A gasoline-powered drill

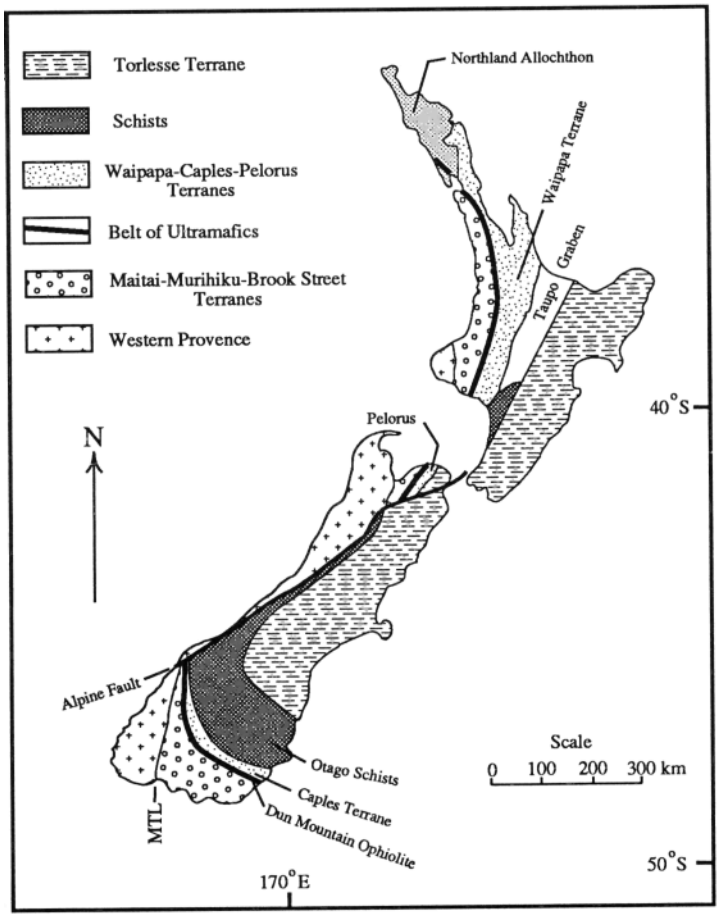


Fig. 1. Generalized terrane map of New Zealand (adapted from Ballance and Sporli [1989]).

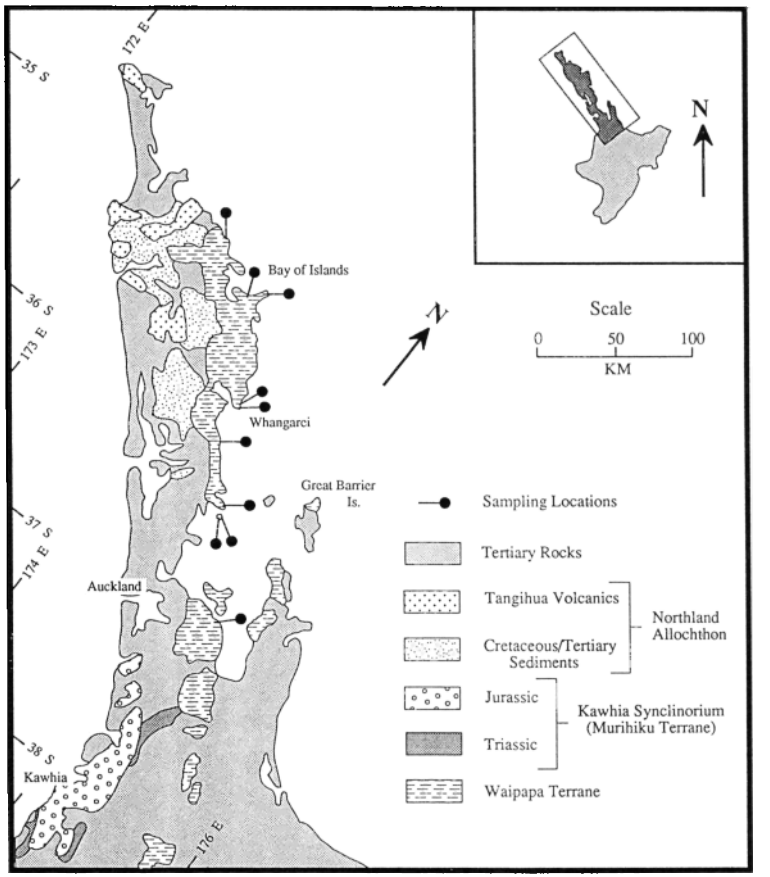


Fig. 2. Generalized geology and paleomagnetic sampling locations from Northland peninsula, North Island, New Zealand (adapted from Sporli [1978]).

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TABLE 1. Paleomagnetic Site Descriptions From the Waipapa Terrane

Site Identification	Comments	Age
KKB86-1	Thin bedded chert; highly fractured, but not cleaved; surrounding area is highly deformed; no reliable magnetic results	Jurassic
KKB86-2	Weathered and fractured red chert ≈200 m from KKB86-1. See KKB86-1.	Jurassic
KI86-1	Very well preserved pillow basalts. Very little deformation, highly veined. Primary mineralogy almost completely altered.	Permian?
KI86-2	Highly deformed and hydrothermally altered thin bedded chert.	Permian?
MB86-1	Thick bedded recrystallized limestone with chert interbeds. Area strongly deformed. Extensive veining in thin section.	Permian
MB86-2A	Thin bed recrytallized limestone on NW limb of shallowly plunging, assymetric anticline. See MB86-1.	Permian
MB86-2B	Thin bed recrytallized limestone on SE limb of fold. See MB86-2a.	Permian
MB86-3	80 cm thick spilitic sill(?). In thin section, ground mass completely altered, some pristine phenochrysts; extensive veining	Permian
MB86-4	Massive recrystallized limestone. See MB86-1.	Permian
OBC86-1	Highly deformed thin bedded chert; near intensely deformed area; no reliable magnetic results.	Permian?
PB86-1	Dark basalt flow overlain by chert beds. Relatively undeformed. Extensive veining in thin section.	Permian?
PB86-2A	Red and green chert beds overlying basalt; site located on north limb of broad fold; no reliable magnetic results.	Permian?
PB86-2B	Red and green chert beds overlying basalt; site located on south limb of broad fold; no reliable magnetic results.	Permian?
RLB86-1	Pillow basalt; extensive hydothermal staining, but well-preserved pillow structures; primary mineralogy nearly completely altered.	Permian?
RLB86-2	Thin (10 cm) red limestone bed overlying pillows of RLB86-1; extensively veined in thin section.	
RLB86-3	Pillow basalt located ≈500 m from RLB86-3. Primary mineralogy much better preserved than in RLB86-1, although extensive veining in thin section.	Permian?
TWB86-1	Black, moderately altered pillow basalts; well preserved pillow forms; veining present in thin section.	Permian?
WBT86-1	Highly fractured, moderately altered pillow basalts; good pillow forms; no reliable magnetic data.	Permian?
WBT86-2	Thin bedded red chert overlying pillow basalts; extensively deformed; veining present in thin section.	Permian?

was used to take 2.5-cm-diameter cores, which were oriented in the field to ±2° by magnetic compass. In this study, a site is defined as a set of samples taken from a single cooling unit or sedimentary horizon. Structural attitudes were taken from sedimentary contacts when possible. In units where structural attitudes were not available, attitudes taken from nearby rocks were used. Each sample was individually stepwise demagnetized using either stepwise alternating field (AF), stepwise thermal, or continuous thermal demagnetization (CTD) [Dunn and Fuller, 1984], and measured on a Molspin spinning magnetometer or a 2G cryogenic magnetometer. Stable directions were selected using equal area and Zijderveld [1967] vector end point plots. Site averages were then computed using the statistics of Fisher [1953].

Of the 19 sites collected, 12 yield stable magnetic directions (Table 2). Figure 3 illustrates typical demagnetization behavior for the samples. The in situ site means cluster together and have a mean direction of I = -70.2°, D=14.4°, α95=7.1°, kappa=38.6 (Figure 4). However, after application of local bedding corrections, the site means scatter widely and the kappa decreases to 3.3 (Figure 5). These sites fail a regional attitude test at the 99% confidence level [MacFadden and Jones, 1981]. In one case, both limbs of a small (5 m), asymmetric, slightly plunging fold were sampled. Before application of a structural correction, the directions are tightly clustered. However, after simple unfolding of the fold, the directions from the two limbs disperse, indicating that the magnetization was acquired after the folding of the beds (Figure 6). The failure of both a regional attitude test and a fold test at the 99% level of confidence [MacFadden and Jones, 1981] indicates that all of the sites sampled from the Waipapa terrane have been remagnetized after the tilting of the rocks.

ROCK MAGNETICS

Standard rock magnetic experiments were carried out on selected samples from the sites which display stable magnetic

TABLE 2. Site Mean Results From the Waipapa Terrane

					In Situ				Bedding	Corrected			
Site	Rock Type	Latitude	Longitude	N	Inc.	Dec.	a95	kappa	Attitude	Inc.	Dec.	a95	kappa
KI86-1	Spilite	36.43° S	174.87°E	5	-78.6	82.3	27.4	8.7	346°/32° W	-52.1	107.4	27.4	8.7
KI86-2	Altered Chert	36.44°S	174.80°E	5	-86.4	39.1	8.9	73.8	203°/63° E	-45.1	111.1	8.9	73.8
MB86-1	Marble	35.00°S	173.79°E	5	-65.1	341.1	5.4	200.1	035°/64° N	-64.5	56.3	5.4	200.1
MB86-2*	Marble	35.00°S	173.79°E	8	-69.1	344.8	3.9	202.1	078°/18°N:020°/68°E	-55.1	315.4	29.5	4.5
MB86-3	Green Spilite	35.00°S	173.79°E	6	-78.8	15.2	10.1	44.8	119°/77° S	-5.8	43.8	10.1	44.8
MB86-4	Marble	35.00°S	173.79°E	6	-62.7	339.8	9.8	47.4	294°/29° N	-60.5	282.8	9.8	47.4
PB86-1	Pillow Basalt	35.27°S	174.17°E	4	-73.1	28.3	9.1	105.3	215°/71° W	-25.4	127.1	9.1	105.3
RLB86-1	Pillow Basalt	35.26°S	174.13°E	5	-61.7	23.6	4.2	329.1	159°/54° W	-20.4	60.3	4.2	329.1
RLB86-2	Limestone	35.26°S	174.13°E	5	-56.4	41.1	9.3	69.1	159°/54° W	-10.4	63.3	9.3	69.1
RLB86-3	Pillow Basalt	35.26°S	174.13°E	6	-64.8	13.6	10.4	42.1	350°/38° W	-44.6	64.4	10.4	42.1
TWB86-1	Pillow Basalt	35.64°S	174.52°E	5	-86.4	39.1	8.9	73.8	008°/44° W	-45.1	111.1	8.9	73.8
WBT86-2	Chert	36.88°S	174.81°E	4	-64.1	8.5	13.9	44.4	357°/76° W	-16.7	81.2	13.9	44
		Mean all sites:		12	-70.2	14.4	7.1	38.6		-47.6	61.6	28.4	3.3

N is number of samples; Inc. is inclination and Dec. is declination, a95 is alpha 95, all three in degrees.

* Fold test: includes data from both limbs of a fold

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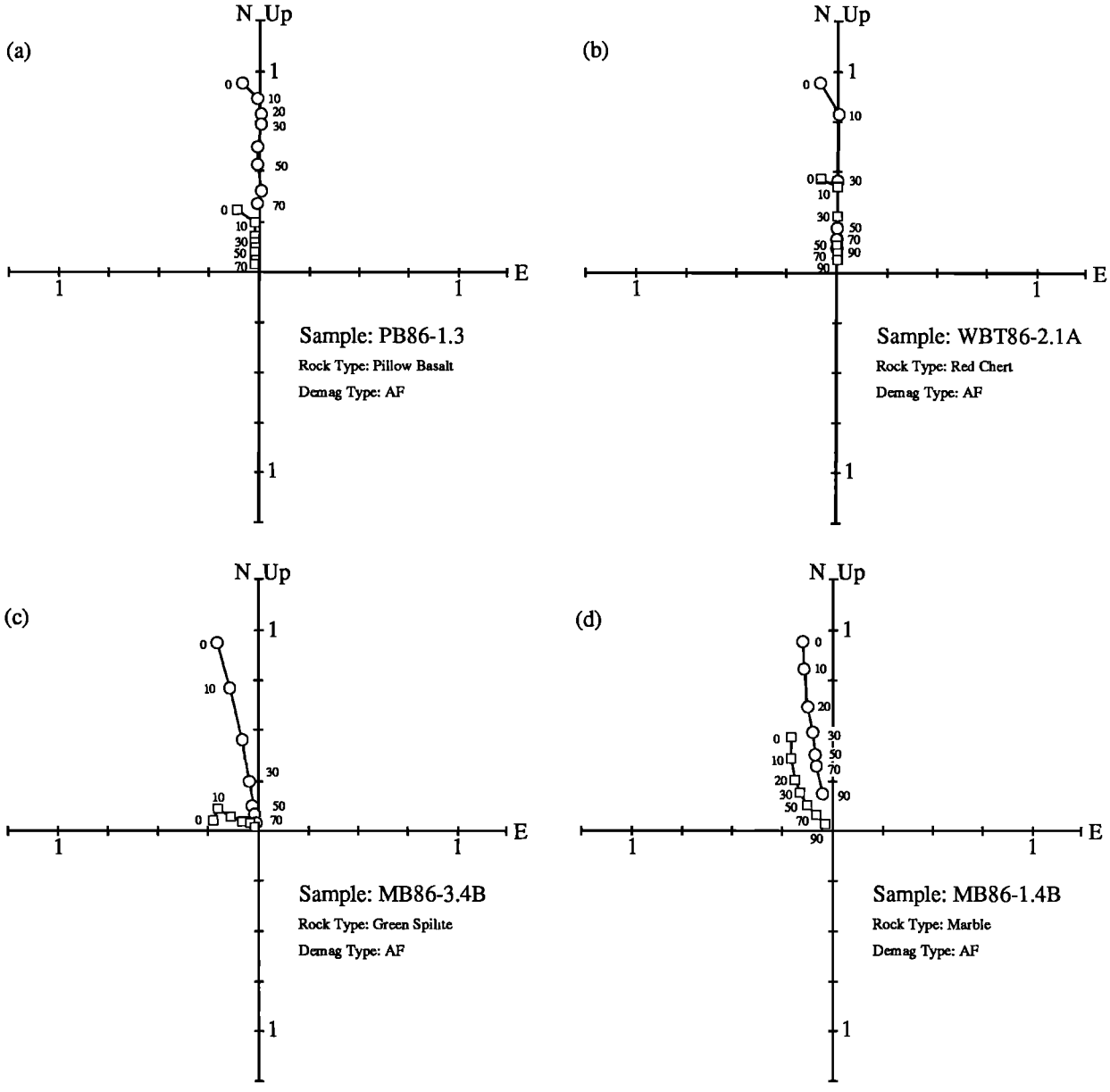


Fig. 3. Orthogonal vector plots of typical demagnetization behavior for Northland sites.

behavior, in an effort to determine the mineralogy and grain size distribution of the material carrying the magnetic remanence in the samples. These experiments included isothermal remanent magnetization (IRM) acquisition, DC and AF demagnetization of IRM, AF and stepwise thermal demagnetization of the natural remanent magnetization (NRM), and continuous thermal demagnetization of saturation IRM (IRM_S). The goal of these experiments is to characterize the nature of the mineral which is carrying the magnetization.

There is wide variation in the magnetic behaviors of the samples from the sites. This variation does not correlate with rock type, structural setting, or geographic location. Table 3 lists the rock magnetic properties for representative samples from the sites which yield stable results. Samples from six of the sites (KI86-2, MB86-1, MB86-4, RLB86-1, RLB86-2, WBT86-2) do not reach IRM saturation (IRM_S) at 800 mT (Figure 7a), and have remanent coercivities (H_{rc}) which range between 220 and 370 mT (Figure 8a). The stepwise thermal demagnetization of these samples indicates that 20-30% of the magnetization remains above 580°C. These data suggest that fine-grained hematite is the dominant carrier of the magnetic remanence in the samples from these sites.

The other six sites (KI86-1, MB86-2, MB86-3, RLB86-3, PB1-5, TWB86-1) all show very different behavior. IRM_S is achieved in all of the samples from these sites (Figure 7b), with the saturation field values (H_{rs}) ranging between 100 and 600 mT. Remanent coercivities for samples from these sites range between 21 and 105 mT (Figure 8b). In all of the samples, the NRM is reduced to zero when they are thermally demagnetized to 580°C. The H_{rc} and H_{rs} values for KI86-1 are

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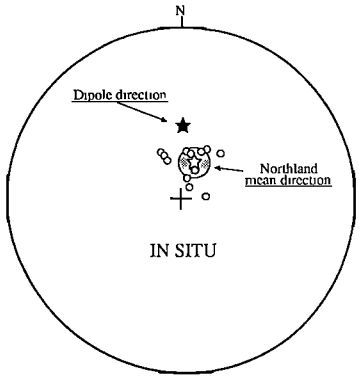


Fig. 4. Equal-area stereographic projection of in situ site mean directions, and the mean direction for all sites with associated 95% confidence interval.

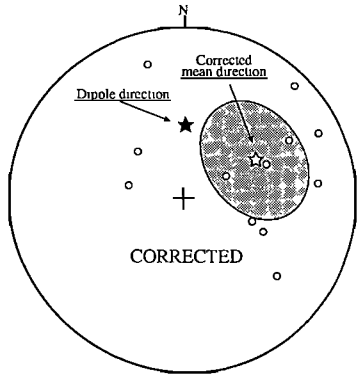


Fig. 5. Equal-area stereographic projection of corrected site mean directions, and the mean direction and associated 95% confidence interval.

higher than would be expected for magnetite (Figure 9). However, continuous thermal demagnetization of IRM_S indicates a Curie temperature around 580°C, which rules out pyrrhotite as the magnetic carrier. The rock magnetic data from these six sites suggest that magnetite is the magnetic carrier, and the relatively high H_{rc} values suggest that the particles are fairly fine grained.

The majority of the sites have NRM intensities which are very low. Site KI86-1 is located in well-preserved pillow basalts, which have an NRM intensity of around 3 x 10^{-7} Am^2/kg, some 3-4 orders of magnitude below what would be considered normal for a lava of this type. Thin section observations indicate that none of the primary mineralogy is preserved and that the rock is composed of secondary alteration products dominated by zeolites. This, coupled with the rock magnetic data, indicates that the primary magnetic mineralogy has been altered to nonmagnetic phases typical of low-temperature alteration.

The rock magnetic data and thin section observations suggest that chemical remanent magnetization (CRM) involving the formation of fine magnetic minerals, and not viscous remanent magnetization (VRM) is probably the primary mechanism for remagnetization in these sites. However, there is evidence which suggests that the alteration of the primary magnetic mineralogy and subsequent formation of fine-grained phases may have been localized. Sites RLB86-1 and 3 come from similar pillow basalts located approximately 400 m apart. The magnetic signal in samples from site RLB86-1 is carried by fine-grained hematite, while the magnetic signal in samples from site RLB86-3 is carried dominantly by multidomain magnetite. Table 3 and Figure 10 illustrate the differences in rock magnetic properties between these two sites which are close in proximity.

DISCUSSION

The 12 sites which yield stable magnetic directions fail both a regional attitude and a fold test [MacFadden and Jones, 1981] at the 99% confidence level, indicating that the rocks were remagnetized after they were folded. The generally low NRM intensities in the rocks and thin section observations suggest that the primary magnetic mineral phases were removed through low-temperature alteration. The presence of fine-grained magnetite and hematite as the main carriers of the magnetic remanence in the majority of the samples suggests that precipitation of fine particles through chemical changes is a likely mechanism for the remagnetization. Extensive veining is observed in all of the samples. Many of the veins show multiple phases of mineral precipitation and contain concentrations of opaque minerals along the boundaries between these phases. This suggests that fluid migration was

TABLE 3. Rock Magnetic Properties for Northland Samples

Sample	J (NRM)	MDF (NRM)	Tbm	Jrs	Hrc	Hrs	MDF (IRM)	Nrm/Irm (0 mT)	Nrm/Irm (50 mT)
KI86-1.6	3.36 e-7*	30	----	6.20 e-5	105	600	100	7.0 e-3	4.0 e-3
KI86-2.6	3.84 e-7	----	670	----	315	----	----	4.0 e-3	4.0 e-3
MB86-1.4B	1.03 e-6	30	640	----	260	----	----	7.0 e-3	5.0 e-3
MB86-2.3A	9.51 e-7	15	----	7.70 e-5	40	300	50	2.0 e-2	8.0 e-3
MB86-3.4B	1.93 e-4	20	580	1.57 e-1	26	200	17.5	2.0 e-3	2.0 e-3
MB86-4.8B	1.76 e-7	40	----	----	290	----	----		
PB1-5	6.27 e-5	22.5	580	4.10 e-4	27.5	200	20	2.0 e-3	2.0 e-3
RLB86-1.2	5.90 e-6	----	580	----	280	----	----	9.0 e-4	9.0 e-4
RLB86-2.5	3.13 e-6	----	640	----	370	----	----	9.0 e-4	9.0 e-4
RLB86-3.3	1.79 e-4	20	560	1.30 e-4	21	100	15	2.0 e-3	9.0 e-4
TWB86-1.1B	9.41 e-7	30	580	1.13 e-4	45	500	50	9.0 e-3	7.6 e-3
WBT86-2.1A	2.43 e-5	22.5	580	----	220	----	---	9.3 e-3	7.6 e-3

J (NRM is intensity of NRM (Am^2/kg); MDF is median destructive field (mT); Tbm is maximum blocking temperature (°C); Jrs is intensity of remanent saturation (Am^2/kg); Hrc is remanent coercivity (mT); Hrs is remanent saturation (mT)

* Read 3.36 e-7 as 3.36 x 10-7

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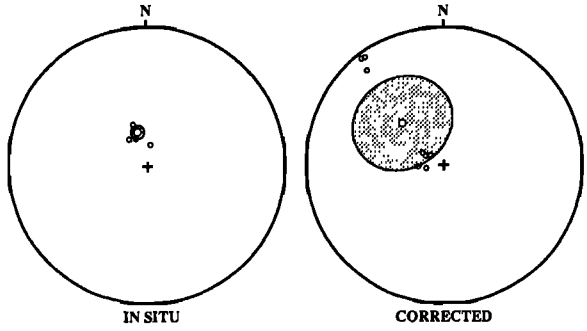


Fig. 6. Equal-area stereographic projection of sample directions with associated site mean direction and 95% interval of confidence from both limbs of a fold at site MB86-2.

prevalent through the rocks and may have been the controling factor in the remagnetization process. The variation of the magnetization over a small distance suggests that the rocks were not affected equally. Faults and fractures may have played a role in the intensity of the alteration of the original magnetic phases.

At the 95% confidence level, the mean paleomagnetic direction is significantly different from the predicted dipole field direction (Figure 4). However, it is not clear when the remagnetization event occurred. The remagnetization event must be younger than the oldest phase of deformation because of the failure of a fold test at the 99% level of confidence (Figure 6); the failure of a regional attitude test suggests that it is probably younger that the youngest deformation event. The

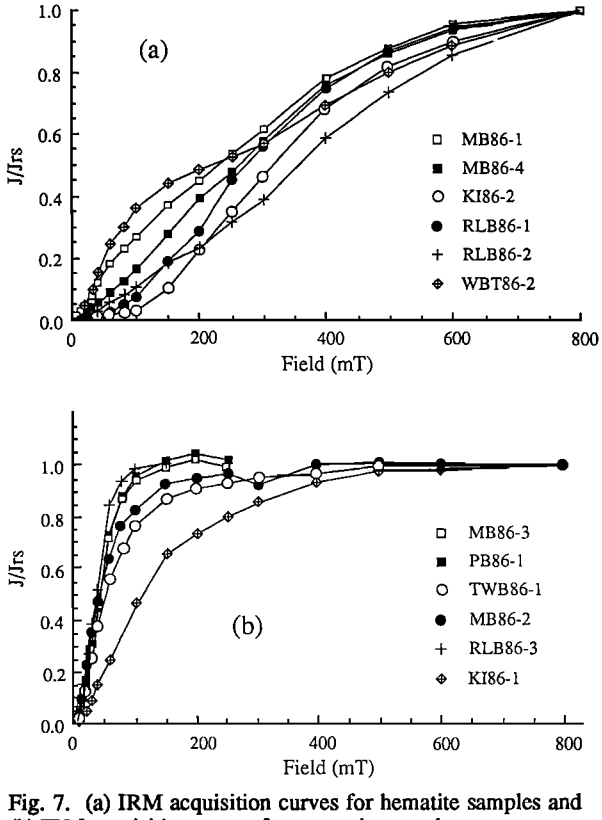


Fig. 7. (a) IRM acquisition curves for hematite samples and (b) IRM acquisition curves for magnetite samples.

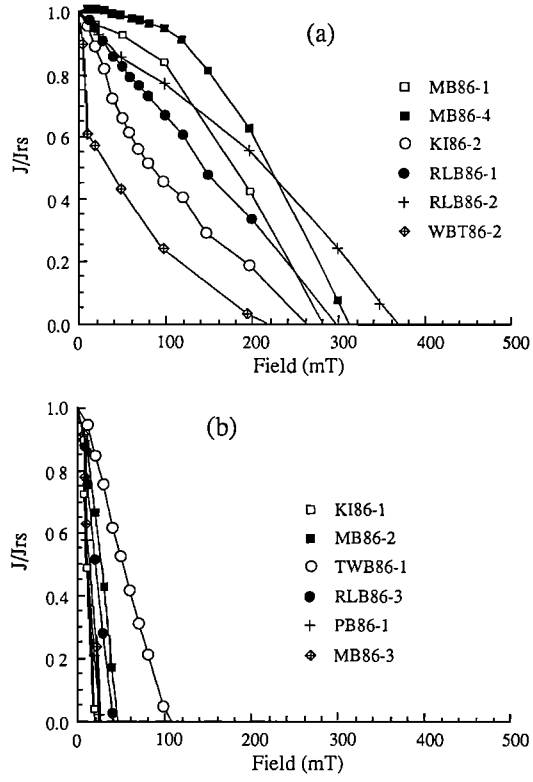


Fig. 8. (a) DC demagnetization of IRM curves for hematite samples and (b) DC demagnetization of IRM curves for magnetite samples.

Waipapa terrane has experienced three phases of deformation [Sporli, 1978]. The first two phases of deformation involved the slicing off and accretion of the subducting oceanic crust along low-angle thrust faults, creating the series of melanges and broken formations found in the Waipapa terrane. These first two phases of deformation are probably associated with the Early Cretaceous Rangitata orogeny [Sporli, 1978].

The third phase of deformation is poorly constrained. The only constraint on the age of this phase of deformation is that it is pre-Cretaceous, and is probably associated with the Kaikoura orogeny. The only expression of this deformation event in the basement rocks of the Waipapa terrane is minor block faulting (K. B. Sporli, personal communication, 1990). The lack of deformation in the Wairakau (Miocene) and Horeke Basalt (Pleistocene) units in the northern part of the study area indicates relative structural quiescence from Miocene time to present [Evans and Hayward, 1988].

Ideally, the remagnetization event can be dated by comparing the observed apparent pole to the New Zealand APW path. Unfortunately, there is no published APW path for New Zealand. In an effort to date the age of the remagnetization event, we have constructed a synthetic APW path for New Zealand by transferring paleomagnetic data from Australia across the Tasman Sea to New Zealand and then comparing it to the apparent pole calculated for the remagnetized Waipapa sites (latitude=-69.7°, longitude=150.7°, α95=11.2°). Data younger than 59 Ma, which corresponds to the cessation of spreading in the Tasman Sea [Molnar et al., 1975; Weissel and Hayes, 1977; Stock and Molnar, 1982], can be used directly for a New Zealand APW path, because there has been no relative movement between Australia and the Northland peninsula after this time. However, data older than 59 Ma was

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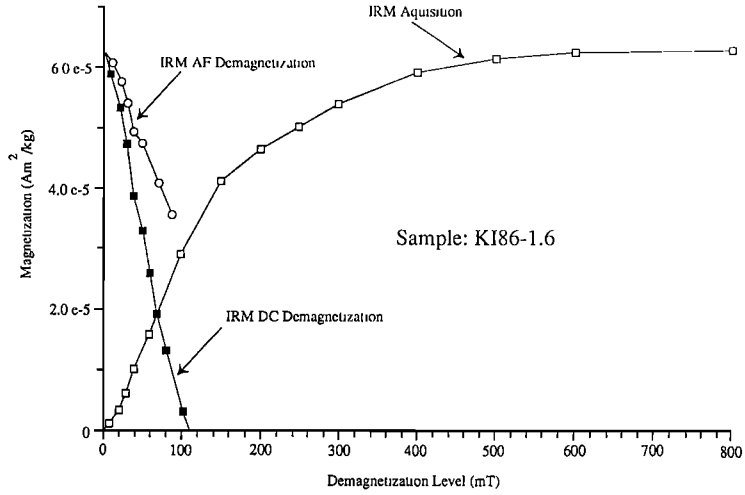


Fig. 9. IRM rock magnetic properties for sample KI86-1.6.

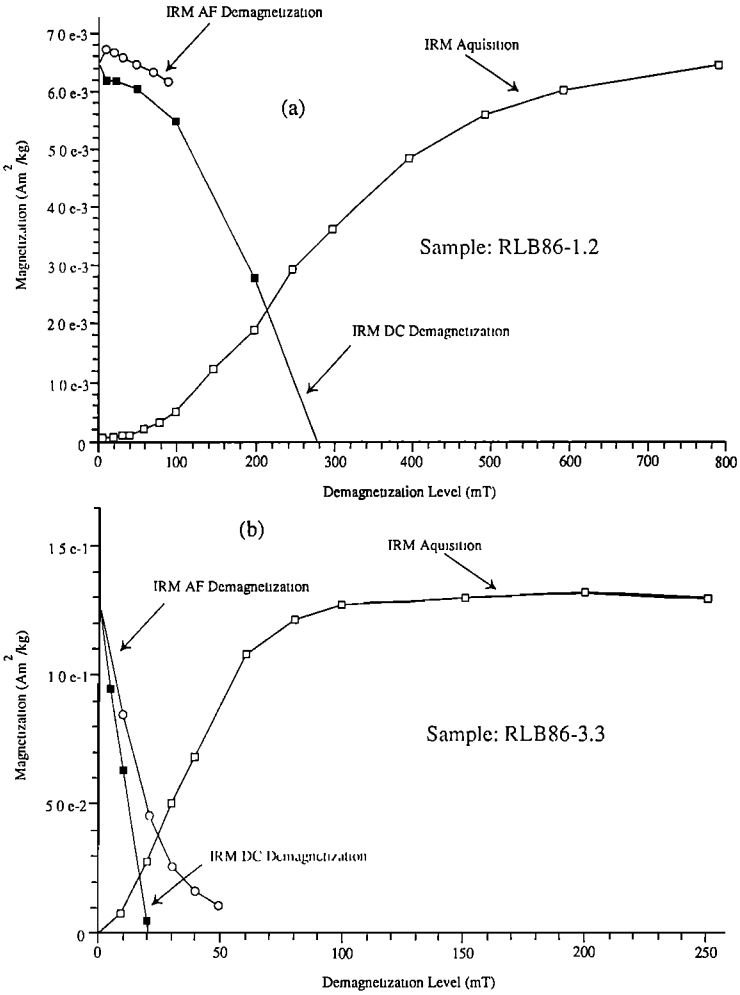
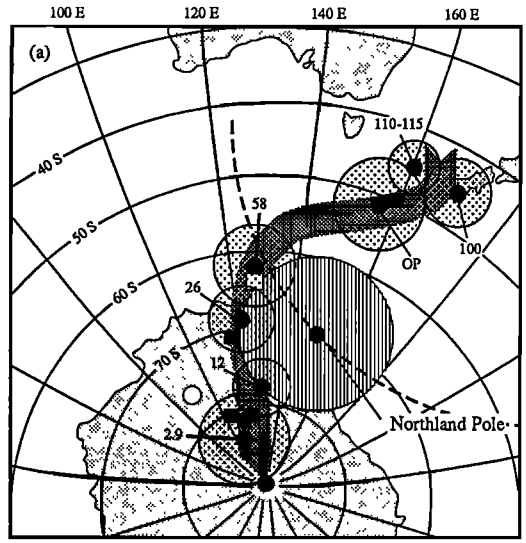


Fig. 10. (a) IRM rock magnetic properties for sample RLB86-1.2. and (b) IRM rock magnetic properties for sample RLB86-3.3.



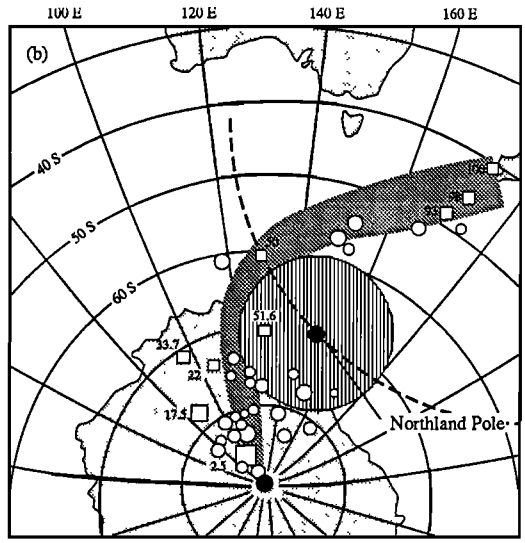


Fig. 11. (a) Synthetic APW path generated from data of Idnurm [1986] using rotation poles from Stock and Molnar (1982). Dated poles are shown in coarse stippling; smoothed path shown in dark gray. Northland pole shown in vertical lines. Solid squares represent poles from laterites. Open square is a best estimate for 50 Ma pole, and open circle is best estimate for 25 Ma pole form Embelton [1981]. OP is pole of inferred Late Cretaceous age. (b) Synthetic APW path generated from data of Musgrave [1989] using rotation poles from Stock and Molnar [1982]. Circular symbols represent laterite and weathered profile poles, squares with their corresponding areas in millions of years represent igneous poles. The stippled area is the trajectory of the APW path within 95% confidence limits.

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transferred across the Tasman Sea using the rotation poles of Stock and Molnar [1982].

There is some uncertainty and disagreement in the published APW paths for Australia. Two different synthetic APW paths were constructed using different data sets, one from Idnurm [1986] and the other from Musgrave [1989]. Figures 11a (from Idnurm's data) and 11b (from Musgrave's data) show the two synthetic APW paths along with the apparent pole from Northland. Data older the 100 Ma have not been included because New Zealand occupied a much higher latitudinal position throughout the Mesozoic [see Grindley et al., 1980; Grindley and Davey, 1982; Haston et al., 1989] than the calculated paleolatitude for the Waipapa terrane.

In Figure 11a, the Northland pole is distinguishable from the axial dipole and the older Late Cretaceous portions of the APW path but intersects the 12, 26, and 58 Ma poles. This suggests that the remagnetization event probably occurred sometime after 58 Ma and before 12 Ma. The APW path of Idnurm [1986] has been questioned, because it implies very rapid polar wander from 26 Ma to present. This is in disagreement with hot spot data from Australia [Wellman, 1983] and requires a significant amount of nondipole field along with some true polar wander to have been present through much of the late Tertiary. In Figure 11b, the Northland pole and its 95% interval of confidence is compared to the synthetic APW path for New Zealand constructed using the data from Musgrave [1989]. As in the previous case, the Northland pole is distinguishable from the axial dipole and the older Late Cretaceous portion of the APW path. The pole intersects the APW path between approximately the 50 and 22 Ma poles. The slower rate of polar wander during the Tertiary suggested by the data of Musgrave [1989] is in better agreement with other geologic evidence, and thus we feel that this data set is the superior of the two.

There is the possibility that a consistent local rotation about a vertical and/or horizontal axis has offset the direction. There are several arguments against this possibility. If there has been a uniform tilting of the Waipapa terrane, either as a whole or in smaller crustal blocks, then this deformation should also be reflected in the younger overlying strata. However, the rocks of the early Miocene Waitemata Group, which overlies much of the Waipapa terrane, are generally structurally simple with gentle dips of 10° or less [Hayward and Brook, 1984]. There is no consistent regional trend to the dip directions in the Waipapa terrane or overlying strata. The orientation of the hinge line axes found in Eocene-Oligocene strata of the Te Kuiti Group changes from a NW to a NNE strike moving from north to south on the Northland peninsula [Schofield, 1983]. These observations suggest that there has been no consistent regional tilting of the basement rocks of the Waipapa terrane.

Any rotation about a vertical axis could also change the position of the calculated pole. The range of possible positions the pole could obtain due to rotation about a local vertical axis is indicated by the the dashed arc on Figure 11. Ballance [1976] originally suggested that the Northland peninsula rotated counterclockwise in the Tertiary. However, work by Walcott [1984] and Lamb [1988] suggests that very little deformation is occurring in the Northland peninsula and that it probably has remained attached to the Australian plate. Even if counterclockwise rotation has occurred in the Northland peninsula, undoing the counter-clockwise rotation has the effect of moving the pole into better agreement with the calculated APW paths. Further, any rotation about a vertical axis will have no affect on the calculated paleolatitude. New Zealand, in general, has been moving southward since the Early Cretaceous, and thus the determination of the age of a paleomagnetic pole by comparing to an APW path will be relatively insensitive to rotation about a vertical axis; in this area, paleolatitude is sufficient to attempt to date the remagnetization event.

It seems unlikely that any local deformation has offset the paleomagnetic directions in a consistent manner over such a large area (>200 km). Considering both of the possible APW paths that can be used, we do not constrain the age of remagnetization very tightly. However, both APW paths imply that the remagnetization event occurred after the Late Cretaceous. The younger bound on the age of the remagnetization is uncertain but must be older than 12 Ma regardless of which APW path is used. If we accept the data of Musgrave [1989], then the youngest bound on the age of remagnetization is early Miocene.

Are there any probable tectonic events which could produce such widespread remagnetizations? The spreading in the Tasman Sea certainly produced an elevated thermal state in the lithosphere around New Zealand [Bradshaw, 1989]. The elevated temperatures associated with rifting lasted between ≈100 and 80 Ma, after which the lithosphere began cooling and subsiding [Sporli and Balance, 1989]. The older limit on the age of the remagnetization event, determined from the APW path, is ≈60 Ma. Therefore elevated temperatures due to initiation of Tasman spreading seem an unlikely mechanism for the remagnetization. However, spreading did not cease in the Tasman Sea until ≈60 Ma [Weissel and Hayes, 1977], and it is possible that the last phase of spreading or the cessation of spreading produced the conditions which remagnetized the rocks in the Waipapa terrane. However, at this time the rocks of the Waipapa terrane were far removed from the axis of spreading.

Between cessation of spreading in the Tasman at ≈60 Ma [Weissel and Hayes, 1977] and the late Oligocene, the rocks in northern New Zealand record a history of stable platform with passive margin subsidence to the east [Sporli and Balance, 1989]. This also seems an unlikely environment to generate such widespread remagnetization, although low-temperature diagenetic changes cannot be completely ruled out. The variable amount of degradation of the primary mineralogy in two closely spaced spilite sites (≈500 m apart; RLB86-1 and RLB86-2) tends to argue against a simple burial metamorphic history.

There was a major change in the tectonic regime at the Oligocene-Miocene boundary (≈25 Ma), which is attributed to the propagation of the India-Pacific plate boundary through New Zealand [Sporli and Balance, 1989]. At this time, carbonate sediments were replaced by the deposition of clastic and volcaniclastic sediments [Nelson and Hume, 1977], the Northland allochthon was emplaced [Ballance and Sporli, 1979; Sporli, 1982], and arc volcanism began along the Northland peninsula [Brothers, 1984; Ballance et al., 1985]. These events mark the beginning of the still ongoing "Kaikoura orogeny". This major change in the tectonic regime seems a likely time for the remagnetizations to have occurred. Fluid migration associated with the obduction of the Northland allochthon, burial of the Waipapa terrane, and initiation of west dipping subduction, and/or elevated temperatures associated with the onset of volcanism are two likely mechanisms for the remagnetization.

Apart from localized block faulting, the Waipapa terrane was generally unaffected by the obduction of the Northland allochthon. However, Oliver [1986] suggests that subduction zones can have a large influence on fluid migration far in front of the down-going slab. Thus, while an area may remain structurally unaffected by a tectonic event, it may see the effects of the event through increased fluid migration. The paleomagnetic samples from Northland all show extensive low-temperature alteration and evidence of fluid migration. Although we cannot completely rule out the other possibilities, it seems that the profound change in the tectonic

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regime in the Northland peninsula in the latest Oligocene/earliest Miocene is the most likely time for this remagnetization event to have occurred. Patterns of remagnetizations due to tectonic events have been well demonstrated in the Appalachian mountains [e.g., Miller and Kent, 1988].

CONCLUSIONS

The paleomagnetic results from the Waipapa terrane indicate that widespread remagnetizations have occurred over a large area (>200 km) throughout the Northland peninsula. The calculated pole for these rocks is distinct from the axial dipole field and suggests that they acquired their magnetization approximately 10° south of their present location. The remagnetization event is constrained to have occurred between ≈12 and 58 Ma, by comparing the computed apparent pole to a synthetic New Zealand APW path. It seems likely that the major change in the tectonic regime of New Zealand at the Oligocene-Miocene boundary, which encompasses the obduction of the Northland allochthon and initiation of subduction and arc volcanism, produced increased fluid flow and/or elevated temperatures which cause the widespread remagnetization event.

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